Submission PHYS1-MID-SUB08

Question PHYS1-MID-Q01
scan faint / margin cut off
visible work:
$v=v$■$+at=4+3(5)=19$ m/s... ?
last line is hard to read; maybe $v=19$ m/s; displacement=57.5 m

Question PHYS1-MID-Q02
scan faint / margin cut off
visible work:
$N=mg=58.8$ N; friction=$\mu N=11.8$ N... ?
last line is hard to read; maybe $a=0.87$ m/s^2; friction opposes motion

Question PHYS1-MID-Q03
scan faint / margin cut off
visible work:
$mgh=\frac{1}{2}mv^2 \Rightarrow v=\sqrt{(2gh)}=\sqrt{(2\cdot 9.8\cdot 8)}$... ?
last line is hard to read; maybe $v=12.52$ m/s

Question PHYS1-MID-Q04
scan faint / margin cut off
visible work:
p_total,before = p_total,after... ?
last line is hard to read; maybe total momentum before equals after when external impulse is negligible

Question PHYS1-MID-Q05
scan faint / margin cut off
visible work:
Impulse is area under $F(t)$: $J=F\Delta t=6(0.5)=3$ N·s.... ?
last line is hard to read; maybe impulse=3 N·s; momentum changes by 3 kg·m/s

Question PHYS1-MID-Q06
scan faint / margin cut off
visible work:
$v=v$■$+at=7+3(2)=13$ m/s... ?
last line is hard to read; maybe $v=13$ m/s; displacement=20 m

Question PHYS1-MID-Q07
scan faint / margin cut off
visible work:
$N=mg=19.6$ N; friction=$\mu N=3.9$ N... ?
last line is hard to read; maybe $a=6.04$ m/s^2; friction opposes motion

Question PHYS1-MID-Q08

scan faint / margin cut off

visible work:

$mgh=\frac{1}{2}mv^2 \Rightarrow v=\sqrt{(2gh)}=\sqrt{(2\cdot 9.8\cdot 9)}$... ?

last line is hard to read; maybe v=13.28 m/s